PYTHON TO MYSQL & PROJECT

Python MySQL

Install MySQL Driver

- Python needs a MySQL driver to access the MySQL database.
- Download and install "MySQL Connector":

C:\Users\Your Name\AppData\Local\Programs\Python\Python36-32\Scripts>python -m pip install mysql-connector-python

Test MySQL Connector

```
import mysql.connector
```

```
(base) C:\Users\A3MAX SOFTWARE TECH>pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.1.0-cp311-cp311-win_amd64.whl.metadata (6.2 kB)
Downloading mysql_connector_python-9.1.0-cp311-cp311-win_amd64.whl (16.1 MB)
   ---------------------------------------- 16.1/16.1 MB 6.4 MB/s eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.1.0

(base) C:\Users\A3MAX SOFTWARE TECH>pip install mysql-connector-python --upgrade
Requirement already satisfied: mysql-connector-python in c:\users\a3max software tech\anaconda3\lib\site-packages (9.1.0)
```

Also user can install from vscode (save **as connection.py**)

connection.py 1 | database.py 1 | database1.py 1

python mysql connector > connection.py > ...

```
1  import mysql.connector
2
3  conn = mysql.connector.connect(host = 'localhost',user='root',password='9876')
4
5  if conn.is_connected():
6      print('connection established')
7  print(conn)
8
9
```

After executing the code you can check the database connection has been established.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| ddlcommand         |
| employee           |
| information_schema |
| joins              |
| mysql              |
| nit                |
| nitdatabase        |
| performance_schema |
| primarykey         |
| pythondb           |
| sys                |
+--------------------+
11 rows in set (0.01 sec)
```

Create another file called as **database.py**

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876')

if conn.is_connected():
    print('connection established')
# print(conn)

mycursor = conn.cursor()
mycursor.execute('create database pythondb')
print(mycursor)
```

Then save another file as **database1.py**

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876')

mycursor = conn.cursor()
mycursor.execute('show databases')
for x in mycursor:
    print(x)
```

```
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE> python -u "c:\Users\A3MAX SOFTWARE TECH\A VS CODE\python mysql connector\database1.py"
('ddlcommand',)
('employee',)
('information_schema',)
('joins',)
('mysql',)
('nit',)
('nitdatabase',)
('performance_schema',)
('primarykey',)
('pythondb',)
('sys',)
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE>
```

Then save another file as '**createtable.py' inside the 'pythondb' create student table**

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876',database='pythondb')

mycursor = conn.cursor()
mycursor.execute('create table student(name varchar(50),branch varchar(10), id int)')
```

Once table created using python you can see the table in the sql as well.

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876',database='pythondb')

mycursor = conn.cursor()
#mycursor.execute('create table student(name varchar(50),branch varchar(10), id int)')
mycursor.execute('show tables')

for x in mycursor:
    print(x)
```

```
python -u "c:\Users\A3MAX SOFTWARE TECH\A VS CODE\python mysql connector\creattable.py"
('student',)
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE>
```

Insert single record to the table – **insertvalue.py**

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876',database='pythondb')

mycursor = conn.cursor()

sql = 'insert into student (name,branch,id) values(%s,%s,%s)'
val = ('john','cse',56)
mycursor.execute(sql,val)
conn.commit()
print(mycursor.rowcount,'record inserted')
```

```
1 record inserted
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE>
```

Insert **multiple record** to the table –

```python
import mysql.connector

conn = mysql.connector.connect(host = 'localhost',user='root',password='9876',database='pythondb')

mycursor = conn.cursor()

sql = 'insert into student (name,branch,id) values(%s,%s,%s)'
#val = ('john','cse',56)

# if user want to create multiple value then you can creat list
val = [('john','cse','56'),('mike','IT','78'),('tyson','me','80')]
#mycursor.execute(sql,val)
mycursor.executemany(sql,val)
conn.commit()
print(mycursor.rowcount,'record inserted')
```

```
TypeError: executemany() missing 1 required positional argument: 'seq_params'
TypeError: executemany() missing 1 required positional argument: 'seq_params'
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE> python -u "c:\Users\A3MAX SOFTWARE TECH\A VS CODE\python mysql connector\insertvalue.py"
3 record inserted
PS C:\Users\A3MAX SOFTWARE TECH\A VS CODE>
```

WORK WITH PYTHON PROJECT WITH MYSQL →

Save the file with **EMPREGISTRATION.PY** file

empregistration.py

python mysql connector > empregistration.py > ...

```python
import tkinter as tk
from tkinter import ttk, messagebox
from tkinter import *

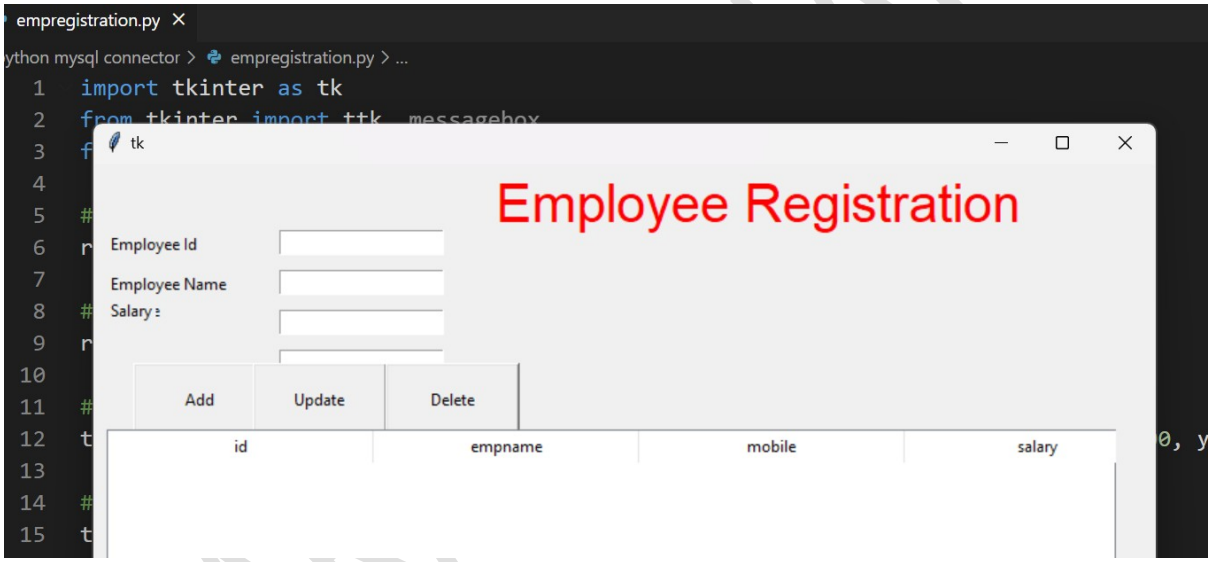
# Create the main window
root = Tk()

# Set the window size
root.geometry('800x500')  # Correct format for geometry

# Create a label for "Employee Registration"
tk.Label(root, text="Employee Registration", fg='red', font=(None, 30)).place(x=300, y=5)

# Labels for Employee Id, Name, Mobile, and Salary
tk.Label(root, text='Employee Id').place(x=10, y=50)
tk.Label(root, text='Employee Name').place(x=10, y=80)
tk.Label(root, text='Mobile').place(x=10, y=100)
```

Once execute above code frontend file will be displayed –



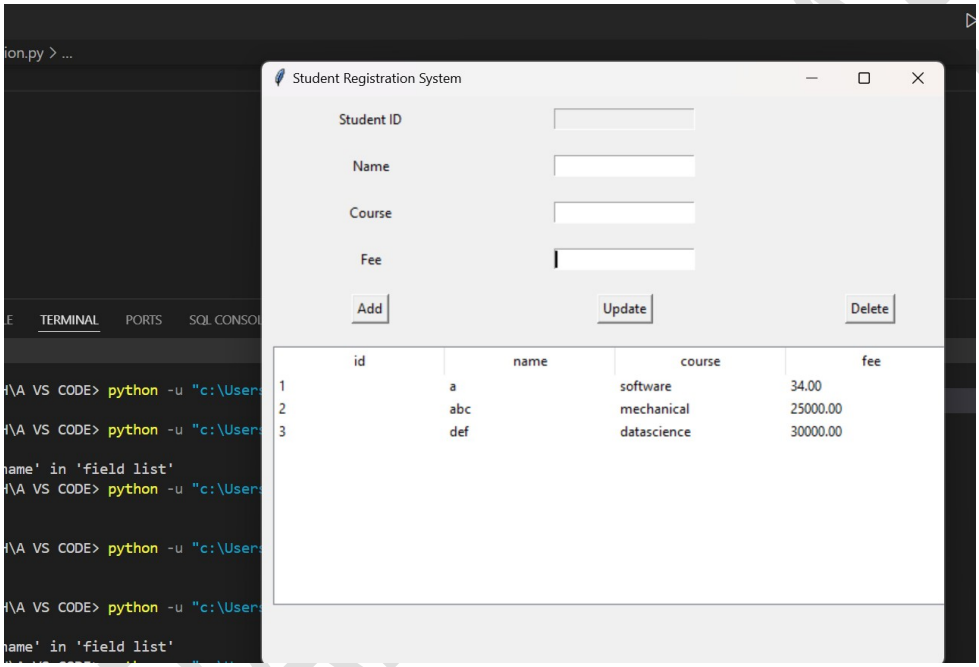
After write the code create new database in mysql tkinter

```
mysql> create database webgui;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use webgui;
Database changed
mysql> CREATE TABLE registration (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL,
    ->     course VARCHAR(100) NOT NULL,
    ->     fee DECIMAL(10, 2) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc registration;
+--------+---------------+------+-----+---------+----------------+
| Field  | Type          | Null | Key | Default | Extra          |
+--------+---------------+------+-----+---------+----------------+
| id     | int           | NO   | PRI | NULL    | auto_increment |
| name   | varchar(100)  | NO   |     | NULL    |                |
| course | varchar(100)  | NO   |     | NULL    |                |
| fee    | decimal(10,2) | NO   |     | NULL    |                |
+--------+---------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

Once you write the backend code then below query will execute



Then you can check to the database – all records will be added.